

Limit Order Book Dynamics and Machine Learning in Algorithmic High-Frequency Trading

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Journal: Journal of Financial Econometrics and Market Microstructure (2024) | Category: Blockchain / FinTech

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ABSTRACT

High-frequency trading firms utilize millisecond limit order book data to predict short-term price fluctuations. We develop microstructural alpha signals using order flow imbalance and volume-weighted average price (VWAP) execution models. Backtesting on global equity and cryptocurrency exchanges demonstrates statistically significant Sharpe ratios under realistic latency and exchange transaction fees.

Keywords: High-Frequency Trading, Limit Order Book, Algorithmic Trading, Financial Markets, Quantitative Finance

1. Introduction

Market microstructure studies the explicit trading mechanics through which prices are established. In continuous double auctions, the limit order book reflects aggregate resting bids and asks.

2. Quantitative Methodology

We extract microstructural features including bid-ask spread depth, queue position, and order cancellation ratios. Machine learning models predict midpoint price changes across microsecond horizons.

3. Empirical Performance

The predictive model attained an information coefficient of 0.12 across major liquidity pairs. Execution strategies reduced market impact costs by 24% relative to standard market orders.

4. Conclusion

Algorithmic execution combining microstructure theory and statistical learning provides critical liquidity and pricing efficiency in modern electronic financial markets.